EX 2 – Demonstrate the statistics

AIM :

To demonstrate basic statistical measures such as mean, standard deviation, variance, correlation, and to understand normal and uniform distributions using a sample dataset in R/Python.

ALGORITHM :

 Import required libraries (NumPy, Pandas, Matplotlib/Seaborn)
 Load or create a sample dataset
 Calculate the mean of the data
 Compute the variance of the dataset
 Calculate the standard deviation
 Analyze the correlation between variables (if multiple features exist)
 Generate and visualize:

- Normal distribution (bell-shaped curve)
- Uniform distribution (equal probability distribution)

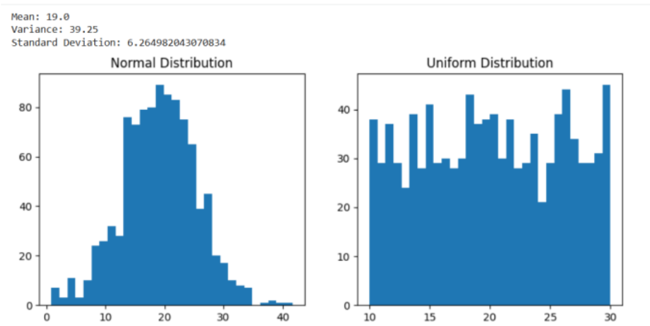
 Plot graphs to visualize the results
 Interpret the statistical measures

CODE :

```
import numpy as np

import matplotlib.pyplot as plt


data=np.array([10,12,15,18,20,22,25,30])


mean=np.mean(data)

variance=np.var(data)

std_dev=np.std(data)


print("Mean:",mean)

print("Variance:",variance)

print("Standard Deviation:",std_dev)
```

```
normal_data=np.random.normal(mean,std_dev,1000)
uniform_data=np.random.uniform(min(data),max(data),1000)

data2=np.array([8,11,14,17,21,23,26,28])

correlation=np.corrcoef(data,data2)

plt.figure(figsize=(10,4))
plt.subplot(1,2,1)
plt.hist(normal_data,bins=30)
plt.title("Normal Distribution")

plt.subplot(1,2,2)
plt.hist(uniform_data,bins=30)
plt.title("Uniform Distribution")

plt.show()
```

OUTPUT :



RESULT :

Thus the above program is executed and the result is obtained.